Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV

Reasoning + Answer:

<think> De Broglie wavelength is given by the formula \(\lambda = \frac{h}{p}\), where \(h\) is Planck's constant and \(p\) is momentum. Momentum is given by \(p = \sqrt{2mE}\), where \(m\) is mass, and \(E\) is energy. For an electron, mass \(m = 9.11 \times 10^{-28} g\). The energy in electron volts is 1 eV = \(1.6 \times 10^{-16} J\). First, convert energy to joules. Next, calculate momentum. Then, use this to find wavelength. </think>

<answer> A. 2.48 ÃĹ10^-8 cm </answer>